What is GRPO?
Quick background of DeepSeekMath & DeepSeekMath training pipeline
PPO vs GRPO: 3 solid differences(problem existed in PPO?)
GRPO step-by-step with a simple example
Reward signal vs reward model
Practical with Huggingface and Unsloth

GRPO

Group Relative Policy Optimization

GRPO was introduced in the DeepSeekMath paper. The authors proposed GRPO as a variant of PPO for reinforcement learning.

- Official website: https://www.deepseek.com/en/
- Chat: https://chat.deepseek.com/
- API platform: https://platform.deepseek.com/
- GitHub: https://github.com/deepseek-ai
- Hugging Face: https://huggingface.co/deepseek-ai

Model Family	Date	Use
DeepSeek-Coder v1 / DeepSeek-LLM	2023	General LLM + coding
DeepSeekMoE	Jan 2024	Efficient MoE model
DeepSeekMath	Feb 2024	Math reasoning
DeepSeek-VL	Mar 2024	Vision-language tasks
DeepSeek-V2	May 2024	Efficient general LLM
DeepSeek-Coder-V2	Jun 2024	Advanced coding
DeepSeek-V2.5	Sep–Dec 2024	Improved chat/coding
DeepSeek-VL2	Dec 2024	Better vision-language
DeepSeek-V3	Dec 2024	Flagship MoE LLM
DeepSeek-R1 / R1-Zero / Distill	Jan 2025	Reasoning model
Janus-Pro	Jan 2025	Multimodal + image generation
DeepSeek-Prover-V2	Apr 2025	Formal theorem proving
DeepSeek-V3.1	Aug 2025	Hybrid thinking/chat
DeepSeek-V3.2 / V3.2-Exp	Sep–Dec 2025	Agentic reasoning
DeepSeek-V4 Preview	Apr 2026	Latest flagship preview

GRPO

DeepSeek-Coder-Base-v1.5 7B ← 2023

↓

Math Continued Pre-training; 120B math tokens + code + natural language

↓

DeepSeekMath-Base 7B ← 2024

↓

Math SFT; CoT + PoT + Tool-integrated reasoning

↓

DeepSeekMath-Instruct 7B

↓

GRPO Reinforcement Learning

↓

DeepSeekMath-RL 7B ← mathatical

DeepSeek-LLM and DeepSeek-Coder v1 are not the same model. DeepSeek-LLM is a general model family, while DeepSeek-Coder is a coding-focused family. DeepSeek-Coder-Base-v1.5 7B is a specific coder checkpoint, and DeepSeekMath was initialized from this checkpoint.

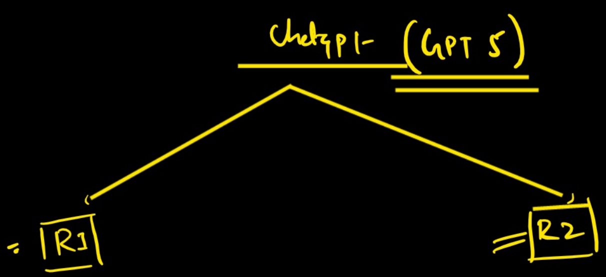
The core idea of the DeepSeekMath paper is to show that a small 7B open-source model can become strong at mathematical reasoning.

They started with DeepSeek-Coder-Base-v1.5 7B because code models already have good logical and structured reasoning ability. Then they continued pre-training it using a large math-focused dataset, including a 120B-token math corpus and other useful data like code, arXiv, and natural language.(DeepSeekMath-Base was trained for 500B tokens, where a major part came from the 120B-token DeepSeekMath math corpus, along with code, arXiv, and natural language data.)

After pre-training, they applied math instruction tuning using step-by-step solution data. Finally, they used GRPO reinforcement learning, where the model generates multiple answers, compares them, and improves based on the better ones.

The final model, DeepSeekMath-RL 7B, achieved 51.7% on the MATH benchmark without external tools or voting. This showed that with good data, instruction tuning, and GRPO, even a 7B open-source model can become very strong in math reasoning.

Its main idea is to remove the separate critic/value model used in PPO and instead calculate the baseline using multiple sampled answers for the same question. This makes RL training more memory-efficient and improve mathematical reasoning.



What is GRPO?

RL

GRPO, or Group Relative Policy Optimization, is a reinforcement learning method used to align LLMs by comparing multiple responses generated for the same prompt.

GRPO improves an LLM by generating multiple responses for the same prompt, scoring them, and updating the model based on which responses are better or worse than the group average, without needing a separate value(critic) model.

GRPO ← light weight algo

Compare PPO

~~Value/critic~~

Prompt - Solve: 2 + 2

Model generates 4 responses:
A: 4 Reward = 1
B: 5 Reward = 0
C: 4 Reward = 1
D: 6 Reward = 0

(value/critic)

Group average reward:
(1 + 0 + 1 + 0) / 4 = 0.5

A = 1, B = 0, C = 1, D = 0

A > 0.5, B < 0.5, C > 0.5, D < 0.5

Now GRPO learns:

Responses above average → increase probability
Responses below average → decrease probability

Instead of using a separate value(critic) model like PPO
GRPO collect a group of responses
scores them
and checks which responses are better or worse than the group average.

Simple flow for GRPO:

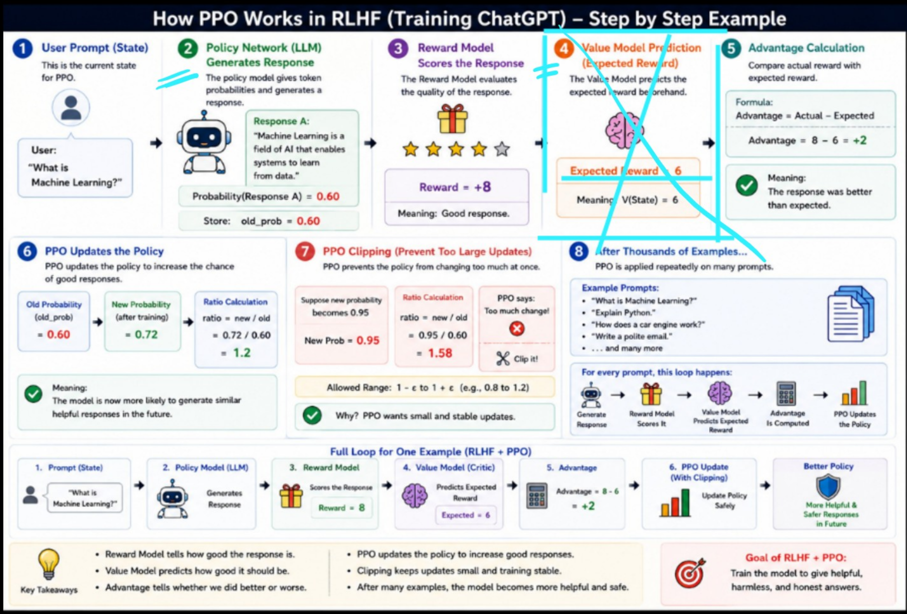
Prompt
↓
Generate multiple responses
↓
Score each response → RS | RM
↓
Compare responses inside the group ← GROUPING
↓
Increase probability of better responses
↓
Decrease probability of weaker responses

So GRPO asks:

"Which answer is better compared to other answers for the same question?"

Instead of PPO-style:

"Was this answer better than what the value model expected?"



DeepSeek used GRPO because PPO is expensive for large LLMs due to the extra value/critic model.
GRPO removes that critic model and compares multiple answers generated for the same question.
The better answers are reinforced, and weaker answers are discouraged.
GRPO trains the LLM by comparing multiple responses within the same group using a reward signal.

Point	PPO	GRPO
1. Value/Critic Model	PPO uses an extra **Value(Critic) model** to estimate expected reward.	GRPO **removes the Value model** and uses group average reward as baseline.
2. Advantage Calculation	PPO advantage = actual reward - value model prediction.	GRPO advantage = response reward compared with other responses in the same group.
3. Memory/Compute Cost	PPO is more memory-heavy because it needs Policy + Reference + Reward + Value model.	GRPO is lighter because it removes the critic/Value model, so training is more memory efficient.

concept

Why GRPO Is Better:

1. No Value/Critic Model
GRPO removes the separate critic/value model used in PPO.
This saves memory and compute.

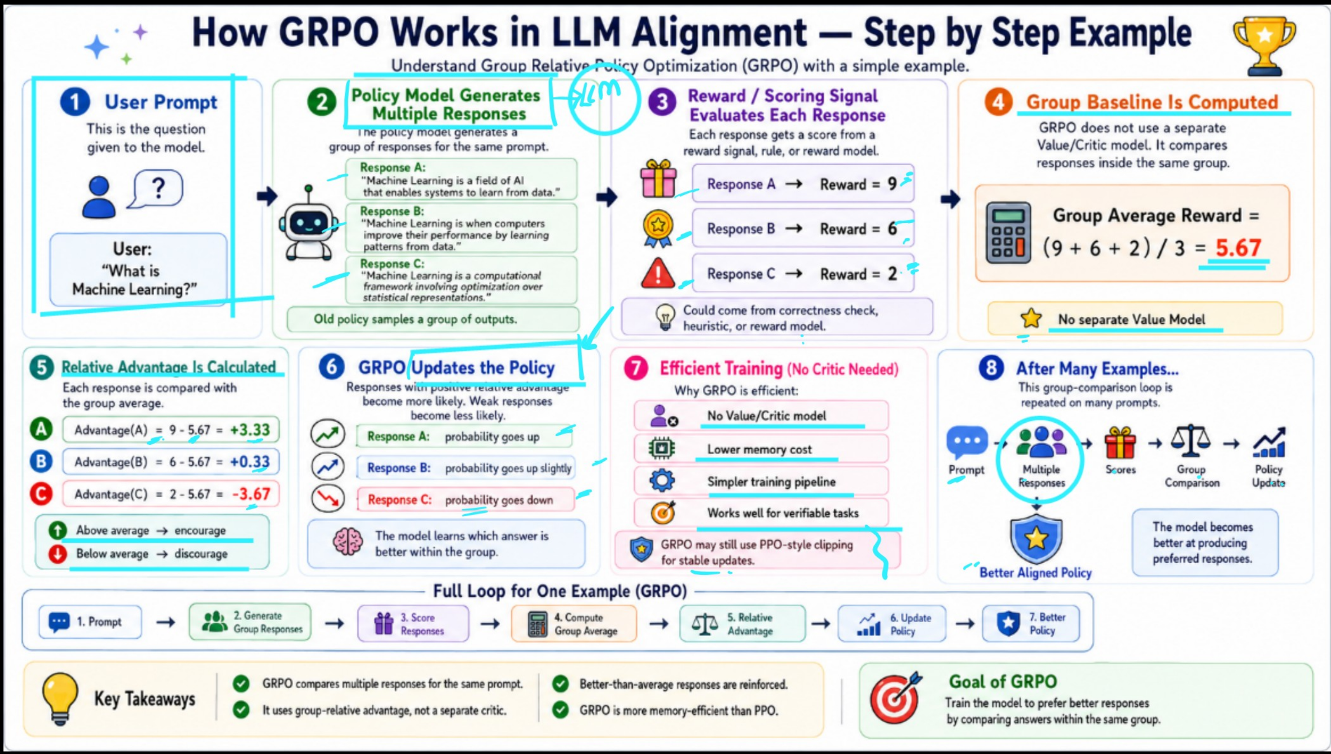
2. Group-Based Comparison
GRPO compares multiple responses for the same prompt.

Example:
Prompt: Solve 2 + 2

A: 4 → good
B: 5 → bad
C: 4 → good
D: 6 → bad

The model learns:
Responses better than group average → increase probability
Responses worse than group average → decrease probability

3. More Efficient for Reasoning Tasks
For math and coding, rewards can be verified using rules.
So we do not always need a human preference reward model.
This makes the training pipeline cheaper and simpler.


How GRPO Works in LLM Alignment — Step by Step Example
Understand Group Relative Policy Optimization (GRPO) with a simple example.
1 User Prompt
This is the question given to the model.
?
User: "What is Machine Learning?"
2 Policy Model Generates Multiple Responses
LLM
The policy model generates a group of responses for the same prompt.
Response A: "Machine Learning is a field of AI that enables systems to learn from data."
Response B: "Machine Learning is when computers improve their performance by learning patterns from data."
Response C: "Machine Learning is a computational framework involving optimization over statistical representations."
Old policy samples a group of outputs.
3 Reward / Scoring Signal Evaluates Each Response
Each response gets a score from a reward signal, rule, or reward model.
Response A → Reward = 9
Response B → Reward = 6
Response C → Reward = 2
Could come from correctness check, heuristic, or reward model.
4 Group Baseline Is Computed
GRPO does not use a separate Value/Critic model. It compares responses inside the same group.
Group Average Reward = (9 + 6 + 2) / 3 = 5.67
No separate Value Model
5 Relative Advantage Is Calculated
Each response is compared with the group average.
A Advantage(A) = 9 - 5.67 = +3.33
B Advantage(B) = 6 - 5.67 = +0.33
C Advantage(C) = 2 - 5.67 = -3.67
Above average → encourage
Below average → discourage
6 GRPO Updates the Policy
Responses with positive relative advantage become more likely. Weak responses become less likely.
Response A: probability goes up
Response B: probability goes up slightly
Response C: probability goes down
The model learns which answer is better within the group.
7 Efficient Training (No Critic Needed)
Why GRPO is efficient:
No Value/Critic model
Lower memory cost
Simpler training pipeline
Works well for verifiable tasks
GRPO may still use PPO-style clipping for stable updates.
8 After Many Examples...
This group-comparison loop is repeated on many prompts.
Prompt
Multiple Responses
Scores
Group Comparison
Policy Update
Better Aligned Policy
The model becomes better at producing preferred responses.
Full Loop for One Example (GRPO)
1. Prompt
2. Generate Group Responses
3. Score Responses
4. Compute Group Average
5. Relative Advantage
6. Update Policy
7. Better Policy
Key Takeaways
GRPO compares multiple responses for the same prompt.
It uses group-relative advantage, not a separate critic.
Better-than-average responses are reinforced.
GRPO is more memory-efficient than PPO.
Goal of GRPO
Train the model to prefer better responses by comparing answers within the same group.

Very Important Point

GRPO does not mean “no reward”.
Wrong: GRPO = no reward
Correct: GRPO still needs a reward signal.

The reward signal can come from:
- Rule-based rewards
- Heuristic rewards
- Task-specific rewards
- Reward model scores

For example:

Math problem → check final answer correctness
Coding problem → run test cases
Reasoning task → reward correct structure or logic

So the correct understanding is:

Reward Model may be removed in some setups.
Reward signal cannot be removed.

Parent term

((Reward Signal))

RB Reward

Reward model

Term	Meaning
Reward Model	A model that judges an output and gives a score.
Reward Signal	The actual score/value used to train the model.

Example:

Prompt: “Explain AI simply.”

Model output A: clear answer
Model output B: confusing answer

A reward model checks both answers and gives scores:

A → 4.8
B → 1.2

These numbers 4.8 and 1.2 are the reward signals.

So simple line:

Reward model is the judge. Reward signal is the score given by the judge.

Important: Reward signal does not always need a reward model. It can also come from rules, humans, tests, or environment feedback. In GRPO/DeepSeekMath style training, rewards are used to compare multiple answers and improve the policy model.

DeepSeek-R1 Example

For reasoning tasks like math and coding, DeepSeek used rule-based rewards.

Examples:

Math → Is the final answer correct?
Coding → Does the solution pass test cases?
Format → Is the reasoning/output in the expected format?

This means the model was not learning blindly from itself.

It was generating multiple answers, and those answers were judged using reward rules or reward signals.

But GRPO does not mean “no reward”. It still needs a scoring signal.
In DeepSeek-style reasoning training, that signal often comes from rule-based rewards like math correctness, code test cases, and format checks.